

Quiz-I Solutions

Q1. $U_1(t) = 0.002 \sin(2\pi \cdot 4t) + 0.001 \sin(2\pi \cdot 50t)$
 $U_2(t) = -0.002 \sin(2\pi \cdot 4t) + 0.001 \sin(2\pi \cdot 50t)$

$$U_o = 4 \sin(2\pi \cdot 4t) + 0.009 \sin(2\pi \cdot 50t)$$

U_1 — $\boxed{\text{Instrumentation amplifier}}$ — U_o
 U_2 —

a) $A_d = \frac{U_{o,d}}{U_{i,d}}$

$$U_{i,d} = U_1 - U_2 = 0.004 \sin(2\pi \cdot 4t)$$

$$U_{o,d} = 4 \sin(2\pi \cdot 4t) \Rightarrow A_d = 10^3 \text{ V/V}$$

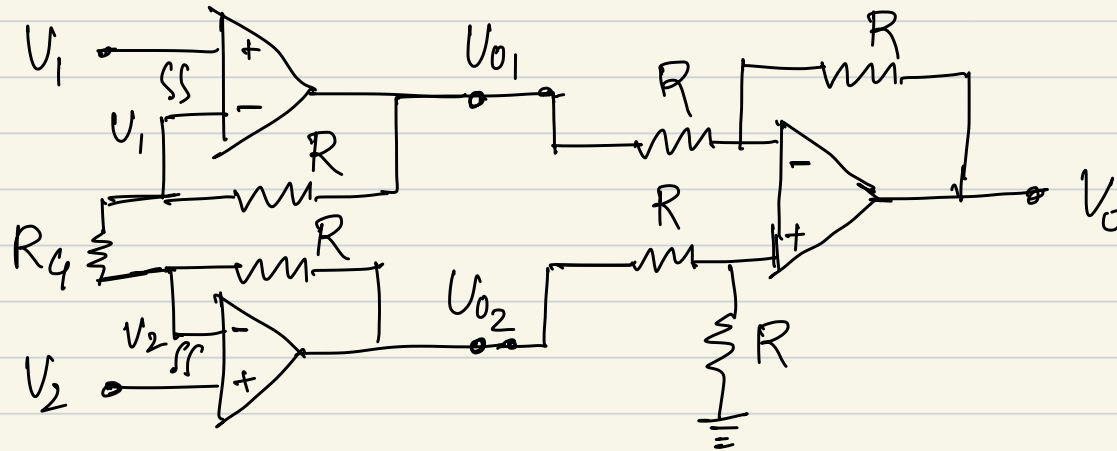
$$U_{c,m} = \frac{U_1 + U_2}{2} = 0.001 \sin(2\pi \cdot 50t)$$

$$\Rightarrow U_{o,c,m} = 0.001 \sin(2\pi \cdot 50t)$$

$$\Rightarrow A_{c,m} = 1 \text{ V/V}$$

b) $CMRR \text{ (in dB)} = 20 \log_{10} \left| \frac{A_d}{A_{cm}} \right| = 20 \log_{10} |1000| = 60 \text{ dB}$

c)



if $R = 500 \text{ k}\Omega$

$R_G \sim 1 \text{ k}\Omega$

or if $R_G = 100 \Omega$
 $R \sim 50 \text{ k}\Omega$

$$V_O = V_{O1} - V_{O2}$$

$$V_{O1} - 2IR - IR_G - V_{O2} = 0 \quad \& \quad V_1 - V_2 = IR_G$$

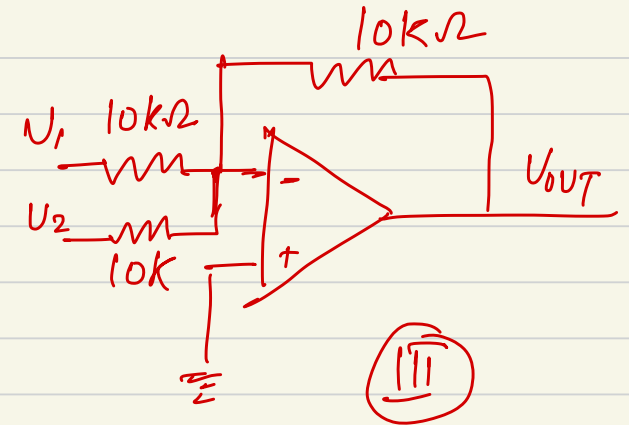
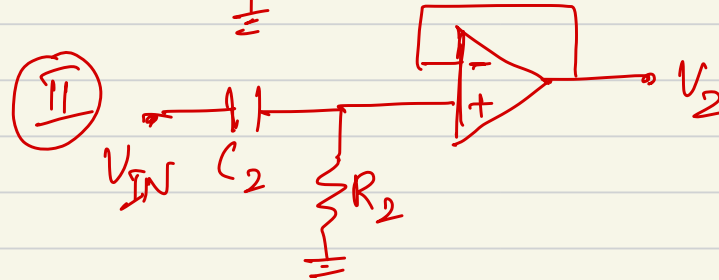
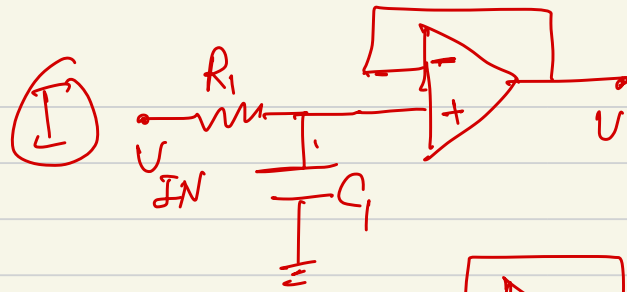
$$\Rightarrow I = \frac{V_{O1} - V_{O2}}{2R + R_G} = \frac{V_1 - V_2}{R_G}$$

$$\Rightarrow V_{O1} - V_{O2} = \left(1 + \frac{2R}{R_G}\right) (V_1 - V_2) = V_O$$

As $A_d = 1000$, $1 + \frac{2R}{R_G} = 1000 \Rightarrow 2R = 999 R_G \Rightarrow$

$$\boxed{R_G = \frac{R}{499.5}}$$

Q2



from (III) circuit:

$$U_{OUT} = - \left(\frac{10K}{10K} U_1 + \frac{10K}{10K} U_2 \right) = - (U_1 + U_2)$$

(I) $U_1 = \frac{1/j\omega C_1}{R_1 + 1/j\omega C_1} \cdot 1 \cdot U_{IN} = \frac{U_{IN}}{1 + j\omega C_1 R_1} = \frac{U_{IN}}{1 + j(\omega/\omega_1)}, \quad \omega_1 = \frac{1}{R_1 C_1}$

(II) $U_2 = \frac{R_2}{R_2 + 1/j\omega C_2} \cdot 1 \cdot U_{IN} = \frac{j\omega C_2 R_2 U_{IN}}{1 + j\omega C_2 R_2} = \frac{j(\omega/\omega_2)}{1 + j(\omega/\omega_2)}, \quad \omega_2 = \frac{1}{R_2 C_2}$

$\Rightarrow \left[\frac{U_{OUT}}{U_{IN}} = - \left(\frac{1}{1 + j\omega C_1 R_1} + \frac{j\omega C_2 R_2}{1 + j\omega C_2 R_2} \right) \right]$

(B)

for A_1 branch to have a cutoff of 15.9 Hz

$$\text{i.e. } f_1 = \frac{1}{2\pi R_1 C_1} = 15.9 \text{ Hz} \Rightarrow R_1 C_1 = \frac{1}{2\pi \times 15.9} \sim 0.01$$

lly for A_2 branch to have a cutoff freq. of 159 Hz

$$\Rightarrow f_2 = \frac{1}{2\pi R_2 C_2} = 159 \text{ Hz} \Rightarrow R_2 C_2 = \frac{1}{2\pi \times 159} \sim 0.001$$

if $C_1 = C_2 = 1 \mu\text{F}$

$$\Rightarrow \boxed{R_1 = 10 \text{ k}\Omega, R_2 = 1 \text{ k}\Omega}$$

(C)

$$\left| \frac{V_{OUT}}{V_{IN}} \right| = \left| \frac{1 + j\omega C_2 R_2 + j\omega C_1 R_2 - \omega^2 C_1 R_1 C_2 R_2}{(1 + j\omega C_1 R_1)(1 + j\omega C_2 R_2)} \right|$$

$$\text{as } \omega_1 = \frac{1}{R_1 C_1}$$

$$\& \omega_2 = \frac{1}{R_2 C_2}$$

$$|A| = \frac{\sqrt{\left(1 - \frac{\omega^2}{\omega_1 \omega_2}\right)^2 + (2\omega/\omega_2)^2}}{\sqrt{1 + (\omega/\omega_1)^2} \sqrt{1 + (\omega/\omega_2)^2}}$$

At $f \rightarrow 0$ or $\omega \rightarrow 0$

$$|A| = \frac{\sqrt{1+0}}{\sqrt{1}\sqrt{1}} \sim 1 \text{ V/V or } 0 \text{ dB}$$

At $f \rightarrow \infty$, or $\omega \rightarrow \infty$

$$|A| = \frac{\sqrt{\frac{\omega^4}{\omega_1^2 \omega_2^2} + 4 \frac{\omega^2}{\omega_2^2}}}{\left(\frac{\omega}{\omega_1}\right) \left(\frac{\omega}{\omega_2}\right)} = \frac{\left(\frac{\omega}{\omega_2}\right) \sqrt{\left(\frac{\omega}{\omega_1}\right)^2 + 4}}{\left(\frac{\omega}{\omega_1}\right) \left(\frac{\omega}{\omega_2}\right)}$$

as $\frac{\omega}{\omega_1} \gg 1$

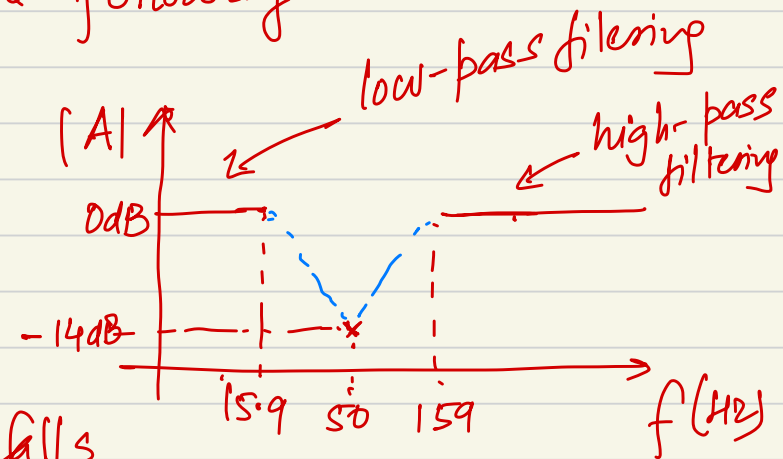
$$|A| \sim 1 \text{ V/V or } 0 \text{ dB}$$

At $f = \sqrt{f_1 f_2}$ i.e. $\omega = \sqrt{\omega_1 \omega_2}$

$$|A| = \frac{\sqrt{0 + \left(\frac{2\omega}{\omega_2}\right)^2}}{\sqrt{1 + \left(\frac{\omega}{\omega_1}\right)^2} \sqrt{1 + \left(\frac{\omega}{\omega_2}\right)^2}} \text{ or } \frac{2f/f_2}{\sqrt{1 + \left(\frac{f}{f_1}\right)^2} \sqrt{1 + \left(\frac{f}{f_2}\right)^2}}$$
$$= \frac{2 \times 50 / 159}{\sqrt{1 + \left(\frac{50}{15.9}\right)^2} \sqrt{1 + \left(\frac{50}{159}\right)^2}} = \frac{0.629}{3.3 \times 1.05} \sim 0.2 \text{ V/V}$$

$$\underline{\text{indB} = -14 \text{ dB}}$$

d) To remove the 50 Hz interference with the given circuit for which the frequency response looks like the following the cut off frequencies of corresponding low-pass filter (Circuit (I)) & cut off freq. of high-pass filter (Circuit (II)) needs to be selected so that the center freq. falls near 50 Hz. That is why, $f_1 = 15.9 \text{ Hz}$ & $f_2 = 159 \text{ Hz}$ are chosen so that $f_c = \sqrt{f_1 f_2}$ at which gain is lowest is near 50 Hz.



Q3.

$$V_i = 4 \sin(2\pi \cdot 4t) \text{ V}$$

for $V_i > 0$, $V_{OA} \rightarrow +ve$

D_1 will conduct as $A_o \rightarrow \infty$

therefore

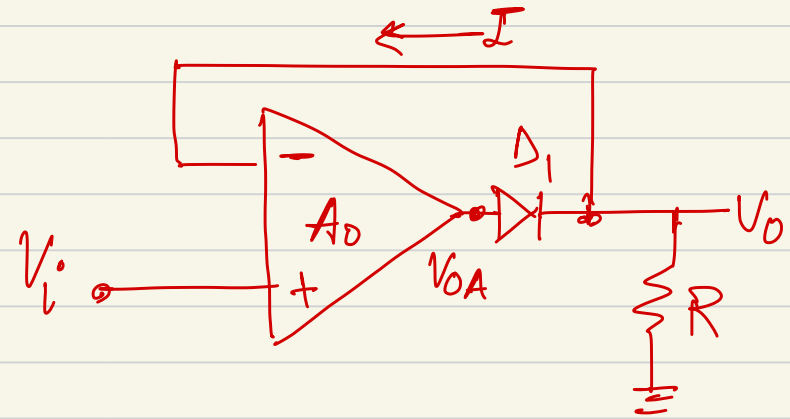
$$A_o(V_+ - V_o) = V_{OA}, V_{OA} \text{ can be } 0.7 \text{ V}$$

$$\Rightarrow V_+ \approx V_o \text{ or } \boxed{V_o \approx V_i}$$

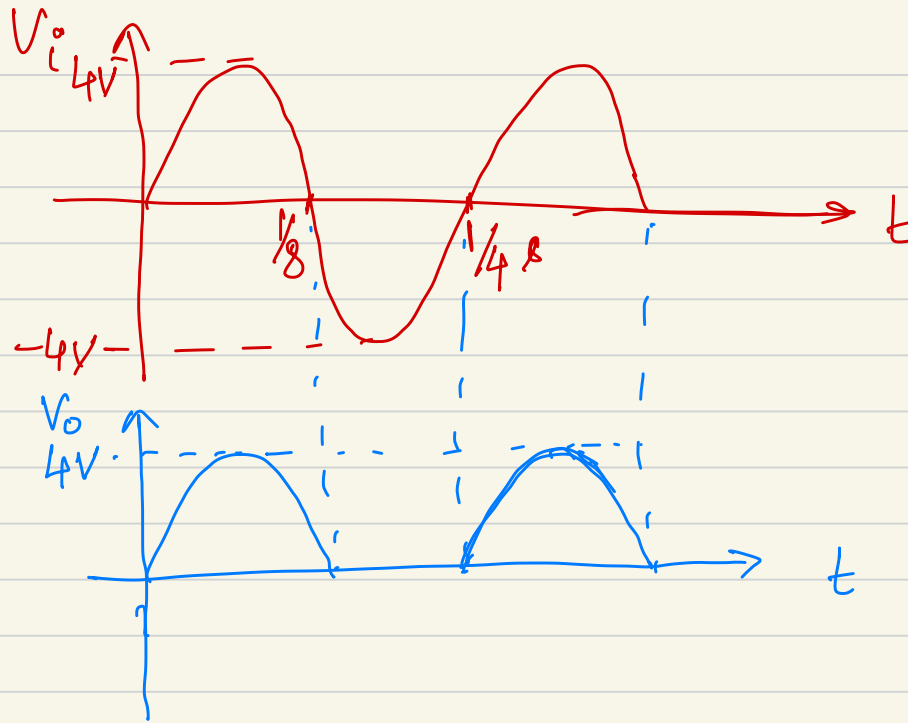
for $V_i < 0$, $V_{OA} \rightarrow -ve$,

$D_1 \rightarrow \text{open}$, it means that the feedback is broken, and as there is no current passing (I) into the Inverting terminal of the Op-Amp,

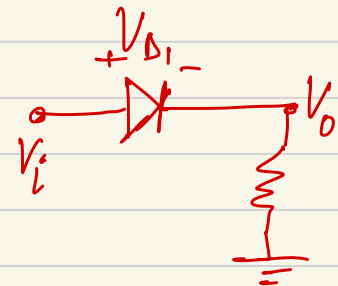
$$\boxed{V_o = 0}$$



$$V_{D_1} = 0.7 \text{ V (F.B.)}$$



(B) If we use a diode & a resistor circuit given here:-



When $V_i > 0$, $V_o = V_i - V_{D1} \Rightarrow V_o = V_i - 0.7V$, while in the above circuit due to a virtual ground, $V_o \approx V_i$ when diode is F.B.

For applications, where we are dealing with biomedical signals, it is important to not lose the voltage across "the diode drop".

Q4. When a signal varying between 0 to 4V is passed through the given circuit (10 bit ADC with $V_{ref} = 5V$), it is required to make a distinction between voltage as small as 2 mV.

(A) In this context, ADC resolution will mean the smallest voltage that it can distinguish, which is at least 2 mV.

(B)* When the switch opens after sampling at $t = t_s$, the capacitor will hold the sampled input value until the next sampling instant when switch closes.

* As ADC conversion time is finite, we need to maintain a constant analog input value at the input of ADC. Therefore, we need a sample & hold circuit to do so.

① Step size for 10 bit ADC with $V_{ref} = 5V$ i.e Full scale range = 5V

$$\Rightarrow LSB = \frac{FSR}{2^N} = \frac{5V}{2^{10}} = 0.00488V \text{ or } 4.88mV$$

d) To get the LSB to be 2mV

$$2^N = \frac{5V}{\underbrace{2mV}} > 2500 \Rightarrow N > \log_2(2500) > 11.2877$$

i.e. $\boxed{N = 12 \text{ bit}}$

i.e we require 12 bit ADC instead of 10 bit one.

Q6.

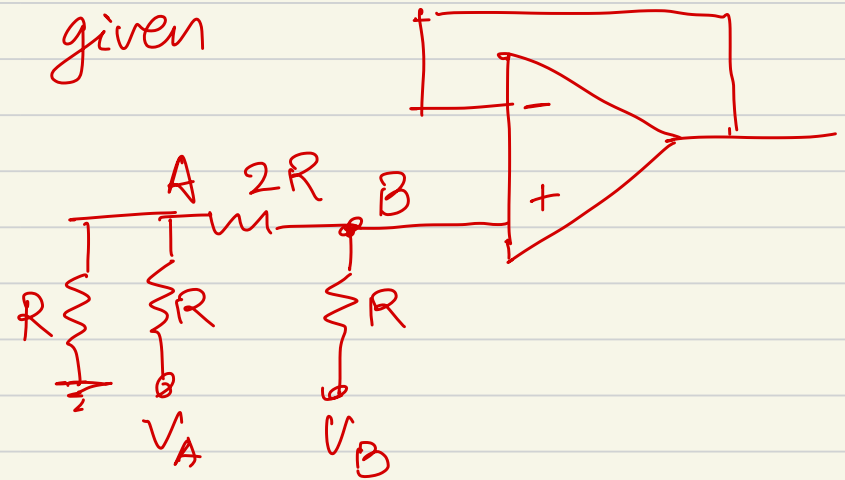
(A)

$$V_{REF} = 5V$$

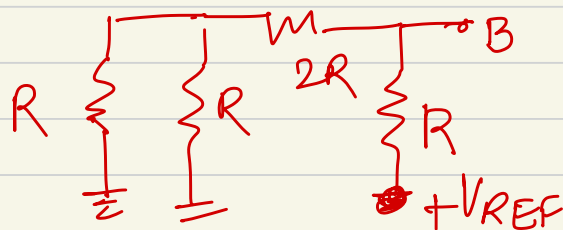
$$V_A = A \cdot V_{REF}$$

$$V_B = B \cdot V_{REF}$$

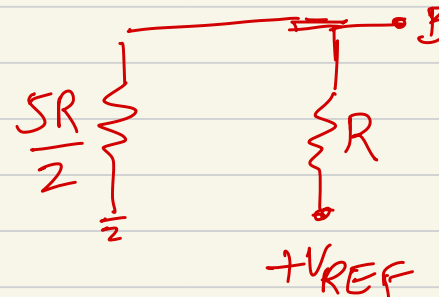
Circuit given



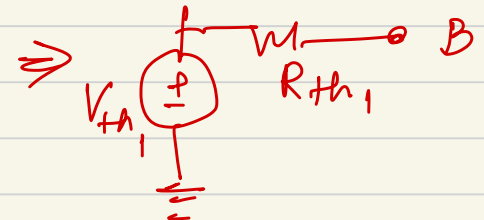
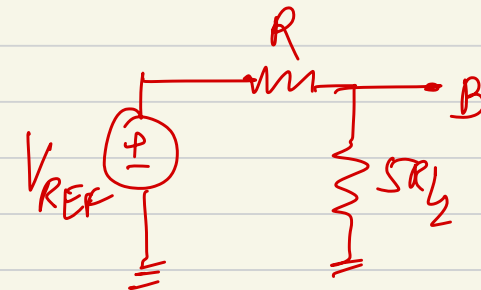
Case 1: $V_B = +V_{REF}$, $V_A = 0$



$\Rightarrow$



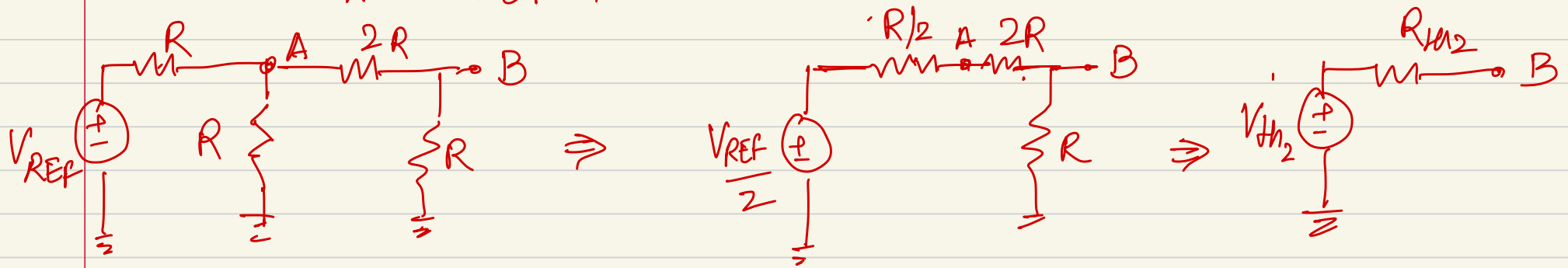
$\Rightarrow$



$$V_{th1} = \frac{5R/2}{7R/2} V_{REF} = \frac{5}{7} V_{REF} \cdot B$$

$$R_{th1} = 5R/7$$

Case 2: $V_A = +V_{REF}$, $V_B = 0$



$$V_{TH2} = \frac{R \times V_{REF}/2}{5R/2 + R} = \frac{V_{REF}}{2} \left[\frac{2}{7} \right] = \frac{V_{REF}}{7} \cdot A$$

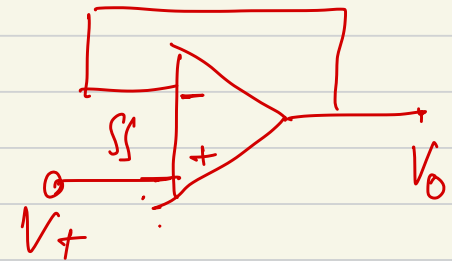
$$R_{TH2} = \frac{5R}{7}$$

Using superposition at node B:-

$$V_+ = V_{TH1} + V_{TH2}$$

$$= \frac{5}{7} V_B + \frac{V_A}{7}$$

$$\Rightarrow \boxed{V_O = V_+} \quad (\text{due to virtual short})$$



(B) if input is $\begin{matrix} V_B & V_A \\ 1 & 0 \end{matrix} \Rightarrow V_B = +V_{REF}, V_A = 0$

$$\Rightarrow V_0 = \frac{5V_{REF}}{7} = \frac{25}{7} = 3.57V$$

if input is $\begin{matrix} V_B & V_A \\ 0 & 1 \end{matrix} \Rightarrow V_B = 0, V_A = +V_{REF}$

$$\Rightarrow V_0 = \frac{5}{7} = 0.714V$$

It means the analog voltage difference between '01' & '10' is

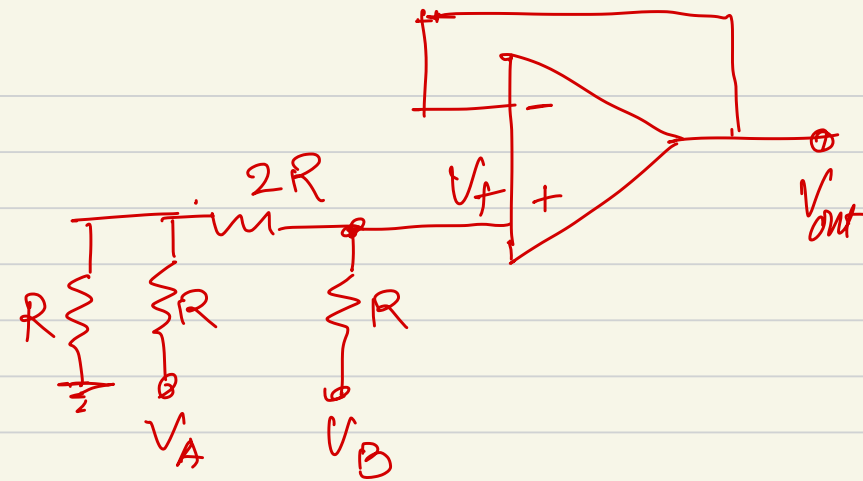
$$= \frac{25}{7} - \frac{5}{7} = \frac{20}{7}V = 2.856V$$

If it is binary weighted, $LSB = \frac{5V}{2^2} = \frac{5}{4}V = 1.25V$

As $\frac{20}{7} \neq \frac{5}{4}$, it is not binary weighted

(c)

As seen from the circuit, V_A & V_B are electrically talking to each other because V_+ is not virtual ground and can take any value dictated by V_A & V_B .



In an inverting summing amplifier, $V_+ = 0$ (physical ground)

i.e. $V_- \approx 0$

* (due to virtual short)

i.e. V_B is isolated from V_A and the current drawn from V_B remains unaffected by V_A .

